

A self-consistent bootstrap for the book algorithm and the bound $R(k, k) \leq 3.769^{k+o(k)}$

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Abstract

Gupta, Ndiaye, Norin and Wei [4] recast the Campos–Griffiths–Morris–Sahasrabudhe book algorithm [2] as an inductive statement, obtaining $R(k, k) \leq (3.7992\dots)^{k+o(k)}$ by two rounds of a bootstrap in which an already established bound on off-diagonal Ramsey numbers enlarges the set of admissible parameters of the next round; they remark that further rounds are possible. We show that the bound *being proved* may itself be used to define the admissible set, so that the whole (infinite) iteration collapses to the verification of a single system of inequalities. Optimizing within this framework—no longer over a closed-form ansatz but over free-form rate functions, discretized as concave piecewise-quadratic splines—gives

$$R(k, \ell) \leq e^{F(\ell/k)k+o(k)} \quad (1 \leq \ell \leq k)$$

for an explicit such F , presented as an exact-integer certificate of 35 770 cells with one tangent witness per cell and orientation, with $e^{F(1)} \leq 3.7689719$; in particular $R(k, k) \leq 3.769^{k+o(k)}$. Self-consistency also removes the hand analysis of small ratios: below $\lambda_0 = 10^{-4}$ the certified F is simply extended linearly, at the cost of a single inequality. The entire proof has been formalized in Lean 4 (Remark 4.4): the final theorems, including the fully formal decimal bound $R(k, k) \leq 3.7690^k$ for all large k , depend only on the axioms `propext`, `Classical.choice`, `Quot.sound`. We also describe how the admissible region must be computed in both orientations $\ell \leq k$ and $\ell > k$, and note that a proposed quintic correction with constant 3.6961 [5] does not satisfy the self-consistency inequality.

1 Introduction

Let $R(k, \ell)$ be the least N such that every red/blue coloring of the edges of K_N contains a red K_k or a blue K_ℓ . Erdős and Szekeres [3] proved $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$, so $R(k, k) \leq 4^k$. After decades of lower-order improvements, Campos, Griffiths, Morris and Sahasrabudhe [2] proved $R(k, k) \leq (4 - 2^{-7})^k$ with their *book algorithm*. Gupta, Ndiaye, Norin and Wei [4] replaced the algorithm by an inductive statement, optimized the parameters and obtained, for $\ell \leq k$,

$$R(k, \ell) \leq e^{G(\ell/k)k+o(k)} \binom{k+\ell}{\ell}, \quad G(\lambda) = (-0.25\lambda + 0.03\lambda^2 + 0.08\lambda^3)e^{-\lambda}, \quad (1)$$

which at $\lambda = 1$ gives $R(k, k) \leq (4e^{-0.14/e})^{k+o(k)} = (3.7992027\dots)^{k+o(k)}$.

Throughout, $H(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$ for $\lambda > 0$, so that $\binom{k+\ell}{\ell} \leq e^{H(\ell/k)k}$ is the Erdős–Szekeres bound in exponential form.

*Revision v3, 2 September 2026. The combinatorial core is the book algorithm of Campos, Griffiths, Morris and Sahasrabudhe [2] in the inductive form of Gupta, Ndiaye, Norin and Wei [4]; Section 3 reproves it, with explicit hypotheses, for the reader’s convenience.

Theorem 1.1. *There is a concave, strictly increasing, piecewise-quadratic C^1 function $F: (0, 1] \rightarrow (0, \infty)$, given explicitly by the certificate of Section 5 (a spline on a partition of $[\lambda_0, 1]$, $\lambda_0 = 10^{-4}$, into 10 560 intervals, extended linearly on $(0, \lambda_0]$), with the following property: for every $\varepsilon > 0$ there is N_ε such that*

$$R(k, \ell) \leq \exp((F(\ell/k) + \varepsilon)k) \quad \text{for all integers } k \geq \ell \geq 1 \text{ with } k + \ell \geq N_\varepsilon.$$

Its terminal value is the exact rational

$$F(1) = \frac{2653604512524788171208898}{2 \cdot 10^{24}} = 1.32680225\dots, \quad e^{F(1)} \leq 3.7689719 < 3.769;$$

in particular

$$R(k, k) \leq 3.769^{k+o(k)}.$$

λ	0.1	0.25	0.5	0.75	0.9	1
$H(\lambda)$	0.3351	0.6255	0.9548	1.1951	1.3143	1.3863
$H(\lambda) + G(\lambda)$ [4]	0.3128	0.5793	0.8896	1.1304	1.2565	1.3348
$F(\lambda)$ (Theorem 1.1)	0.3023	0.5680	0.8823	1.1230	1.2487	1.3268

Table 1: Exponents in $R(k, \lambda k) \leq e^{(\cdot)^{k+o(k)}}$. The values of F are evaluated from the certificate nodes (Section 5.2).

The improvement over (1) at the diagonal is the factor $(3.76897/3.79920)^k = e^{-0.00799k}$; Table 1 lists some off-diagonal values.

Context and what is new

The proof in [4] has two parts. The combinatorial part (their Lemma 12 and Theorem 13) says, roughly, that a candidate pair (X, Y) with red density $\geq p$ and $|X||Y| \geq x^{-k}y^{-\ell}\mu^{-\ell}$ contains one of the desired cliques, provided $x < (1 - \mu)p^{1/(1-\mu)}$ and (x, y) is *admissible*, i.e. $x^{-k'}y^{-\ell'}$ is (asymptotically) an upper bound for $R(k', \ell')$ for all k', ℓ' . The analytic part (their Theorem 14) converts this into a bound $R(k, \ell) \leq e^{F(\ell/k)k+o(k)}$ for any smooth F such that, with $p = 1 - e^{-F'}$ and $X = (1 - M)p^{1/(1-M)}$, some admissible $(X(\lambda), Y(\lambda))$ satisfies

$$F(\lambda) > -\frac{1}{2}(\log X(\lambda) + \lambda \log M(\lambda) + \lambda \log Y(\lambda)) \quad (0 < \lambda \leq 1). \quad (2)$$

The larger the admissible $Y(\lambda)$, the smaller F can be; and the admissible set is enlarged by any already established bound on Ramsey numbers. In [4] this is exploited in two rounds: the Erdős–Szekeres bound gives a first $F_{0.08}$, which enlarges the admissible set and gives $F_{0.03}$, i.e. (1). The authors remark that additional rounds are possible and report that an unverified third round would give 3.78233 [4, Remark 17].

Our contribution is threefold.

1. *Self-consistency* (Section 4). We prove that the admissible set may be computed from the very function F one is trying to establish (Theorem 4.2). A direct induction is circular at its base; we resolve this with a *shift ladder*: the functions $F + s$, $s \downarrow 0$, are shown valid one after another, each step being an application of the one-round theorem to the previous one. This replaces the infinite iteration by a single system of inequalities and yields the

fixed point of the bootstrap. The ladder theorem—and, as of this revision, the entire proof of Theorem 1.1—is formalized in Lean 4 on top of the RamseyLean development of [4] (Remark 4.4).

2. *Optimization and certification* (Section 5). Instead of optimizing within a closed-form ansatz (as the previous revision of this paper did, with a degree-8 correction to the Erdős–Szekeres exponent, reaching 3.772), we compute the fixed point of the bootstrap in a discretization of the framework: F is a free-form concave piecewise-quadratic spline on a 10 560-interval partition of $[\lambda_0, 1]$, $\lambda_0 = 10^{-4}$, determined interval by interval by the minimal admissible slope, with a per-interval book parameter M and pair (X, Y) , admissibility being tested against F itself—self-consistently, and in both orientations. The maximal admissible Y is controlled by a Legendre-type transform of the symmetric extension of F (Lemma 5.1, equivalent to [4, Lemma 15]), so that one tangent-line witness per cell and orientation suffices. The proof object is a certificate of 35 770 cells whose entries are exact scaled integers; self-consistency permits a *linear* extension of F below λ_0 , in place of the hand analysis of small ratios that a closed-form ansatz requires. The certificate is verified by a Python checker and, bit-identically, by the Lean kernel.
3. *Both orientations* (Section 6). The admissible region must dominate the Ramsey bound at all ratios $\ell/k \in (0, \infty)$, i.e. in both orientations; in our certificate every cell carries one tangent witness per orientation. We also note that the self-consistent framework confirms the constant 3.7992... of [4] directly, without any chain of iterations, and rules out a proposed quintic with constant 3.6961 [5].

2 Notation, valid bounds and admissible pairs

All colorings are red/blue edge colorings of a complete graph. $N_R(v)$ and $N_B(v)$ are the red and blue neighborhoods of a vertex v . For disjoint non-empty vertex sets X, Y , $e_R(X, Y)$ is the number of red edges between them and $d(X, Y) = e_R(X, Y)/(|X||Y|)$. A *candidate* is a pair (X, Y) of disjoint non-empty sets; it is (k, ℓ, t) -*good* if $X \cup Y$ contains a red K_k , or X contains a blue K_t , or Y contains a blue K_ℓ . As K_1 is a vertex, every candidate is (k, ℓ, t) -good when $k = 1$ or $t = 1$. A *blue book* (S, T) consists of disjoint S, T with S a blue clique and all S – T edges blue.

Definition 2.1 (valid functions). $G: (0, 1] \rightarrow \mathbb{R}$ is *valid* if for every $\varepsilon > 0$ there is N such that $R(k, \ell) \leq \exp((G(\ell/k) + \varepsilon)k)$ for all integers $k \geq \ell \geq 1$ with $k + \ell \geq N$. Its *symmetric extension* is $\tilde{G}(t) = G(t)$ for $0 < t \leq 1$ and $\tilde{G}(t) = tG(1/t)$ for $t > 1$.

By $R(k, \ell) = R(\ell, k)$, if G is valid then for all $k, \ell \geq 1$ with $k + \ell \geq N$, writing $t = \ell/k$,

$$R(k, \ell) \leq \exp(\tilde{G}(t)k + \varepsilon \max(k, \ell)) \quad (3)$$

(for $\ell > k$: $R(\ell, k) \leq e^{(G(k/\ell) + \varepsilon)\ell}$ and $G(k/\ell)\ell = \tilde{G}(t)k$). The function H is valid with $N = 1$, $\varepsilon = 0$: $\binom{k+\ell}{\ell} \leq (k+\ell)^{k+\ell} k^{-k} \ell^{-\ell} = e^{H(\ell/k)k}$.

Definition 2.2 (admissible pairs). For $G: (0, 1] \rightarrow \mathbb{R}$, a pair $(x, y) \in (0, 1)^2$ is G -*admissible* if $-\log x - t \log y \geq \tilde{G}(t)$ for all $t > 0$.

Thus (x, y) is G -admissible iff $x^{-k}y^{-\ell} \geq e^{\tilde{G}(\ell/k)k}$ for all $k, \ell \geq 1$, i.e. iff $x^{-k}y^{-\ell}$ dominates the bound (3) (up to the o -term) for *every* ratio, in both orientations. The set $\{(x, y) : G\text{-admissible}\}$ is closed under decreasing x or y .

3 The key lemma

Lemma 3.4 is [4, Lemma 12] (a compression of the book algorithm of [2]) with the hypotheses on Ramsey numbers made explicit. We include the complete proof, following [4].

Lemma 3.1 (Cauchy–Schwarz). *For every candidate (X, Y) : $\sum_{v \in X} e_R(X, N_R(v) \cap Y) = \sum_{y \in Y} \deg_R(y, X)^2 \geq e_R(X, Y)^2 / |Y|$.*

Proof. Both sums count triples (v, u, y) with $v, u \in X$, $y \in Y$, vy and uy red; then apply Cauchy–Schwarz to $\sum_y \deg_R(y, X)$. $\square$

Lemma 3.2 (books). *Let $0 < \mu < 1$, let $b, m, k \geq 1$ be integers with $m \geq 5b^2/\mu$, and let X be a vertex set with $|X| \geq 5m^2$ such that at least $R(k, m)$ vertices $v \in X$ satisfy $|N_B(v) \cap X| \geq \mu|X|$. Then X contains a red K_k or a blue book (S, T) with $|S| = b$ and $|T| \geq \frac{1}{2}\mu^b|X|$.*

Proof. Let $W = \{v \in X : |N_B(v) \cap X| \geq \mu|X|\}$. Since $|W| \geq R(k, m)$, W contains a red K_k (done) or a blue K_m with vertex set U . For $z \in X \setminus U$ let $\sigma_z = |N_B(z) \cap U|/m$. Then $m \sum_{z \in X \setminus U} \sigma_z = e_B(U, X \setminus U) \geq \sum_{u \in U} (|N_B(u) \cap X| - (m-1)) \geq m(\mu|X| - m)$, so the mean $\bar{\sigma}$ of the σ_z satisfies $\bar{\sigma} \geq \mu - m/|X|$. Pick $S \subseteq U$ uniformly at random with $|S| = b$ and let $T = \{z \in X \setminus U : S \subseteq N_B(z)\}$, so that (S, T) is a blue book. If $\sigma_z m \geq b$ then

$$\Pr[z \in T] = \binom{\sigma_z m}{b} / \binom{m}{b} = \prod_{i=0}^{b-1} \frac{\sigma_z m - i}{m - i} \geq \left(\frac{\sigma_z m - b}{m - b} \right)^b \geq \left(\sigma_z - \frac{b}{m} \right)_+^b,$$

since $(a - i)/(m - i)$ is non-increasing in i when $a \leq m$. By convexity of $u \mapsto (u - b/m)_+^b$,

$$\mathbb{E}|T| \geq \sum_{z \in X \setminus U} \left(\sigma_z - \frac{b}{m} \right)_+^b \geq |X \setminus U| \left(\bar{\sigma} - \frac{b}{m} \right)_+^b \geq (|X| - m) \left(\mu - \frac{m}{|X|} - \frac{b}{m} \right)_+^b.$$

Here $b/m \leq \mu/(5b)$ and $m/|X| \leq 1/(5m) \leq \mu/(25b^2) \leq \mu/(25b)$, so the last factor is at least $\mu^b(1 - \frac{6}{25b})^b \geq 0.76\mu^b$ (Bernoulli), and $|X| - m \geq (1 - \frac{1}{5m})|X| \geq 0.8|X|$. Hence $\mathbb{E}|T| \geq 0.608\mu^b|X|$. $\square$

Lemma 3.3. *Let $0 < \eta < 1/2$, $p, \mu \in [\eta, 1 - \eta]$ and let $r \geq 2\eta^{-1} \log(1/\eta)$ be an integer. Then*

$$(p^{1/r} - \mu)^r (1 - \mu)^{1-r} \geq (1 - \mu) p^{1/(1-\mu)} \left(1 - \frac{\log^2(1/\eta)}{\eta^2 r} \right).$$

Proof. Let $u = \log p/(1 - \mu)$, so $|u| \leq \eta^{-1} \log(1/\eta) \leq r/2$. From $p^{1/r} \geq 1 + (\log p)/r$ we get $(p^{1/r} - \mu)/(1 - \mu) \geq 1 + u/r > 0$. For $|v| \leq 1/2$, $\log(1 + v) \geq v - v^2$, so $(1 + u/r)^r \geq e^{u - u^2/r} \geq e^u(1 - u^2/r)$. Multiply by $1 - \mu$. $\square$

Lemma 3.4 (key lemma). *Let $0 < \eta < 1/4$, $0 < \varepsilon < \eta/4$, let $r \geq 1$ be an integer and let $x, y, \mu, p \in (0, 1)$ satisfy $\mu \geq \eta$, $p - \varepsilon \geq \eta$, $x + \varepsilon < 1$, $y + \varepsilon < 1$, $\mu + \varepsilon < 1$ and*

$$x^{1/r} (1 - \mu - \varepsilon)^{1-1/r} + \mu + 2\varepsilon \leq (p - \varepsilon)^{1/r}, \quad (4)$$

$$\varepsilon(\mu + x) \leq \mu(1 - \mu - x). \quad (5)$$

There is $L_0 = L_0(\varepsilon, r, \eta)$ such that for all integers $\ell \geq L_0$, $k \geq 1$, $t \geq 1$ the following holds. Suppose

(H1) $R(k', \ell) \leq (x + \varepsilon)^{-k'}(y + \varepsilon)^{-\ell}$ for every integer $1 \leq k' \leq k$;

(H2) (X, Y) is a candidate such that, with $\delta_s := \varepsilon/s$ and $\alpha_0 := d(X, Y) + \delta_{k+t} - p$,

$$\alpha_0 > 0 \quad \text{and} \quad \alpha_0^r |X| |Y| \geq x^{-k} y^{-\ell} \mu^{-t}. \quad (6)$$

Then (X, Y) is (k, ℓ, t) -good.

Choice of L_0 . Write $n = k + t$, $N = k + \ell + t$ and

$$b = \left\lceil \frac{r \log((1 + \varepsilon)n^2/\varepsilon) + \log 2}{\log(1 + \varepsilon/\mu)} \right\rceil, \quad m = \lceil 5b^2/\mu \rceil, \quad w = (n + m)^m. \quad (7)$$

Since $\mu \geq \eta$, we have $\log(1 + \varepsilon/\mu) \geq \log(1 + \varepsilon)$ and $m \leq \lceil 5b^2/\eta \rceil$, so $b = O(\log n)$, $m = O(\log^2 n)$ and $\log(2n^3w/(\varepsilon\eta)) = O(\log^3 n)$ with constants depending only on ε, r, η . Hence there are K_1, C_1 depending only on (ε, r, η) such that $(1 + \varepsilon)^n \geq \max\{5m^2, 2n^3w/(\varepsilon\eta), 1/\varepsilon\}$ for $n \geq K_1$ and $\max\{\dots\} \leq C_1$ for $n < K_1$. Let $(1 + \varepsilon)^{L_0-r} \geq C_1$. Then, for $\ell \geq L_0$ and all $k, t \geq 1$ (using $p - \varepsilon \geq \eta$),

$$(1 + \varepsilon)^{N-r} \geq \max\left\{5m^2, \frac{2n^3w}{\varepsilon(p - \varepsilon)}, \frac{1}{\varepsilon}\right\}. \quad (8)$$

□

Proof of Lemma 3.4. Fix $\ell \geq L_0$ and induct on $n = k + t$; the cases $k = 1, t = 1$ are trivial, so let $k, t \geq 2$ and assume the statement for smaller $k + t$ (same ℓ and parameters).

Step 0. If $|Y| \geq (x + \varepsilon)^{-k}(y + \varepsilon)^{-\ell}$ then $|Y| \geq R(k, \ell)$ by (H1), so Y contains a red K_k or a blue K_ℓ and we are done. Hence assume $|Y| < (x + \varepsilon)^{-k}(y + \varepsilon)^{-\ell}$. As $\alpha_0 \leq 1 + \varepsilon$, (6) gives

$$|X| \geq \frac{x^{-k} y^{-\ell} \mu^{-t}}{\alpha_0^r |Y|} > \frac{(1 + \varepsilon/x)^k (1 + \varepsilon/y)^\ell \mu^{-t}}{(1 + \varepsilon)^r} \geq (1 + \varepsilon)^{N-r}, \quad (9)$$

and also $\alpha_0 |X| \geq (1 + \varepsilon)^{N-r}$ (if $\alpha_0 \leq 1$ then $\alpha_0 |X| \geq \alpha_0^r |X| > (1 + \varepsilon)^N$ by the same computation; if $\alpha_0 > 1$ then $\alpha_0 |X| > |X|$). By (8),

$$|X| \geq 5m^2, \quad w \leq \frac{\varepsilon(p - \varepsilon)}{2n^3} |X|, \quad \frac{1}{\alpha_0 |X|} < \varepsilon. \quad (10)$$

Step 1: regularisation. If some $v \in X$ has $|N_R(v) \cap Y| < (p - \delta_n)|Y| = (d(X, Y) - \alpha_0)|Y|$, put $X' = X \setminus \{v\}$. Then $e_R(X', Y) > d(X, Y)|X||Y| - (d(X, Y) - \alpha_0)|Y|$, so $d(X', Y) > d(X, Y) + \alpha_0/(|X| - 1)$ and $\alpha'_0 := d(X', Y) + \delta_n - p > \alpha_0 |X|/(|X| - 1)$, whence $(\alpha'_0)^r |X'| |Y| > \alpha_0^r |X| |Y|$. Thus (X', Y) satisfies (H2) (and $X' \neq \emptyset$), and goodness of (X', Y) implies that of (X, Y) . Repeating, we may assume

$$|N_R(v) \cap Y| \geq (p - \delta_n)|Y| \geq (p - \varepsilon)|Y| \quad (v \in X); \quad (11)$$

(9)–(10) remain valid since they only used (6) and $|Y|$.

Step 2: many vertices of large blue degree. Let $W = \{v \in X : |N_B(v) \cap X| \geq (\mu + \varepsilon)|X|\}$ and note $R(k, m) \leq \binom{k+m}{m} \leq w$. Suppose $|W| \geq w$. By Lemma 3.2 (with $\mu + \varepsilon$ for μ ; $m \geq 5b^2/(\mu + \varepsilon)$ and $|X| \geq 5m^2$), X contains a red K_k (done) or a blue book (S, T) with $|S| = b$, $|T| \geq \frac{1}{2}(\mu + \varepsilon)^b |X|$.

If $t \leq b$ then $S \supseteq$ a blue K_t (done). Otherwise put $t' = t - b \geq 1$ and consider (T, Y) with parameters (k, ℓ, t') . By (11), $d(T, Y) \geq p - \delta_n$, so

$$\alpha_0(T, Y) := d(T, Y) + \delta_{k+t'} - p \geq \delta_{n-b} - \delta_n = \frac{\varepsilon b}{n(n-b)} \geq \frac{\varepsilon}{n^2} > 0,$$

and by (6), $\alpha_0 \leq 1 + \varepsilon$ and the choice (7) of b ,

$$\alpha_0(T, Y)^r |T| |Y| \geq \left(\frac{\varepsilon}{n^2}\right)^r \frac{(\mu + \varepsilon)^b}{2} \cdot \frac{x^{-k} y^{-\ell} \mu^{-t}}{(1 + \varepsilon)^r} = \frac{(1 + \varepsilon/\mu)^b}{2} \left(\frac{\varepsilon}{(1 + \varepsilon)n^2}\right)^r x^{-k} y^{-\ell} \mu^{-t'} \geq x^{-k} y^{-\ell} \mu^{-t'}.$$

(H1) is unchanged. By induction (T, Y) is (k, ℓ, t') -good; a blue $K_{t'}$ in T together with S is a blue K_t in X , and the other outcomes are inherited. So (X, Y) is good.

Step 3: choosing a vertex. Now let $|W| < w$. For $v \in X$ let $Y_v = N_R(v) \cap Y \neq \emptyset$ and $d_v = d(X, Y_v)$; put $d = d(X, Y)$. Lemma 3.1 gives $\sum_v |Y_v| d_v \geq d^2 |X| |Y| = d \sum_v |Y_v|$, i.e. $\sum_{v \in X} |Y_v| (d_v - d) \geq 0$; the terms with $v \in W$ contribute less than $w|Y|$. If $d_v < d - \gamma$ for all $v \in X \setminus W$, where $\gamma = \varepsilon/n^3$, then by (11) $-w|Y| < -\gamma(|X| - w)(p - \varepsilon)|Y|$, so $w > \gamma(p - \varepsilon)|X|/2$, contradicting (10). Fix therefore $v \in X \setminus W$ with

$$d(X, Y') \geq d(X, Y) - \frac{\varepsilon}{n^3}, \quad |Y'| \geq (p - \varepsilon)|Y|, \quad |N_B(v) \cap X| < (\mu + \varepsilon)|X|, \quad Y' := N_R(v) \cap Y, \quad (12)$$

and let $X_R = N_R(v) \cap X$, $X_B = N_B(v) \cap X$, $p' = p - \delta_{n-1}$, $\alpha = d(X, Y') - p'$, $\alpha_R = d(X_R, Y') - p'$, $\alpha_B = d(X_B, Y') - p'$ (the last two when the sets are non-empty). Since $\delta_{n-1} - \delta_n = \varepsilon/(n(n-1)) \geq \varepsilon/n^3$, (12) gives $\alpha \geq \alpha_0 > 0$.

Step 4a: red step. Suppose $X_R \neq \emptyset$, $\alpha_R > 0$ and $\alpha_R^r |X_R| \geq \frac{x}{p - \varepsilon} \alpha^r |X|$. Consider (X_R, Y') with parameters $(k - 1, \ell, t)$: (H1) holds a fortiori, $\alpha_0(X_R, Y') = \alpha_R > 0$, and

$$\alpha_R^r |X_R| |Y'| \geq \frac{x}{p - \varepsilon} \alpha^r |X| (p - \varepsilon) |Y| \geq x \alpha_0^r |X| |Y| \geq x^{-(k-1)} y^{-\ell} \mu^{-t}.$$

By induction (X_R, Y') is $(k - 1, \ell, t)$ -good; as v is red to all of $X_R \cup Y'$, a red K_{k-1} there gives a red K_k , and the other outcomes are inherited. So (X, Y) is good.

Step 4b: blue step. Suppose $X_B \neq \emptyset$, $\alpha_B > 0$ and $\alpha_B^r |X_B| \geq \frac{\mu}{p - \varepsilon} \alpha^r |X|$. Consider (X_B, Y') with parameters $(k, \ell, t - 1)$: (H1) holds, $\alpha_0(X_B, Y') = \alpha_B > 0$ and $\alpha_B^r |X_B| |Y'| \geq \mu \alpha_0^r |X| |Y| \geq x^{-k} y^{-\ell} \mu^{-(t-1)}$. By induction it is $(k, \ell, t - 1)$ -good; a blue K_{t-1} in X_B with v is a blue K_t in X . So (X, Y) is good.

Step 4c: neither applies—contradiction. Splitting $e_R(X, Y') - p' |X| |Y'|$ along $X = X_R \sqcup X_B \sqcup \{v\}$ and using $e_R(\{v\}, Y') = |Y'|$ gives $\alpha |X| |Y'| = \alpha_R |X_R| |Y'| + \alpha_B |X_B| |Y'| + (1 - p') |Y'|$ (terms for empty sets being 0), hence

$$1 \leq \frac{(\alpha_R |X_R|)_+}{\alpha |X|} + \frac{(\alpha_B |X_B|)_+}{\alpha |X|} + \frac{1}{\alpha |X|}. \quad (13)$$

As Step 4a does not apply, either $(\alpha_R |X_R|)_+ = 0$ or $\alpha_R |X_R| = (\alpha_R^r |X_R|)^{1/r} |X_R|^{1-1/r} < (\frac{x}{p - \varepsilon})^{1/r} \alpha |X|^{1/r} |X_R|^{1-1/r}$; similarly for B . With $a = |X_B|/|X| \in [0, \mu + \varepsilon)$ and $|X_R|/|X| \leq 1 - a$, (13) yields

$$(p - \varepsilon)^{1/r} \leq f(a) + \frac{(p - \varepsilon)^{1/r}}{\alpha |X|}, \quad f(a) := x^{1/r} (1 - a)^{1-1/r} + \mu^{1/r} a^{1-1/r}. \quad (14)$$

For $r \geq 2$, $f'(a) = (1 - \frac{1}{r})[(\mu/a)^{1/r} - (x/(1-a))^{1/r}] \geq 0$ iff $a \leq \mu/(\mu+x)$, and (5) says $\mu + \varepsilon \leq \mu/(\mu+x)$; so f is non-decreasing on $[0, \mu + \varepsilon]$ (constant if $r = 1$). Using $\mu^{1/r}(\mu + \varepsilon)^{1-1/r} \leq \mu + \varepsilon$ (weighted AM-GM), $f(a) \leq f(\mu + \varepsilon) \leq x^{1/r}(1 - \mu - \varepsilon)^{1-1/r} + \mu + \varepsilon$. Finally $\alpha|X| \geq \alpha_0|X| > 1/\varepsilon$ by (10), so (14) gives $(p - \varepsilon)^{1/r} < x^{1/r}(1 - \mu - \varepsilon)^{1-1/r} + \mu + 2\varepsilon$, contradicting (4). $\square$

4 The bootstrap

4.1 One round

The following is [4, Theorem 14] with two harmless modifications: the identity defining X is relaxed to an inequality (the proof only ever uses the inequality), and admissibility is phrased through a valid function G rather than through the set $\mathcal{R}$ of [4]. We include the proof to make all constants explicit.

Theorem 4.1 (one round). *Let $G: (0, 1] \rightarrow \mathbb{R}$ be valid. Let $F: (0, 1] \rightarrow [0, \infty)$ be C^1 with $F' > 0$ and $C_F := \sup_{0 < \lambda \leq 1} (H(\lambda) - F(\lambda))/\lambda < \infty$, and let $M, X, Y: (0, 1] \rightarrow (0, 1)$ be continuous with*

$$X(\lambda) \leq (1 - M(\lambda))p(\lambda)^{1/(1-M(\lambda))}, \quad p(\lambda) := 1 - e^{-F'(\lambda)}. \quad (15)$$

Assume that for every $\lambda \in (0, 1]$:

(A1) $(X(\lambda), Y(\lambda))$ is G -admissible;

(A2) $\psi(\lambda) := F(\lambda) + \frac{1}{2}(\log X(\lambda) + \lambda \log M(\lambda) + \lambda \log Y(\lambda)) > 0$.

Then F is valid.

Proof. Fix $\varepsilon \in (0, 1/4]$. We shall find K such that $R(k, \ell) \leq e^{(F(\ell/k) + 2\varepsilon)k}$ for all $k \geq K$ and $1 \leq \ell \leq k$; since $k \geq (k + \ell)/2$, this gives validity with $N = 2K$.

Small ratios. Put $\varepsilon_0 = \min\{1, \varepsilon/(2C_F)\}$ if $C_F > 0$ and $\varepsilon_0 = 1$ otherwise. For $\ell \leq \varepsilon_0 k$, $R(k, \ell) \leq e^{H(\lambda)k} \leq e^{(F(\lambda) + C_F \lambda)k} \leq e^{(F(\lambda) + \varepsilon/2)k}$, $\lambda = \ell/k$. Let $I = [\varepsilon_0, 1]$.

Constants. By continuity and compactness choose $\eta \in (0, 1/4)$ with $X, Y, M, p \in [\eta, 1 - \eta]$ on I , let $\psi_{\min} = \min_I \psi > 0$, and put $\theta = \min\{2\varepsilon, \psi_{\min}/2\} \leq 1/2$. By uniform continuity of F' on I choose $\theta' \in (0, \min_I F'/2)$ so that $p_\lambda := 1 - e^{-F'(\lambda) + \theta'}$ satisfies $p_\lambda \geq \eta/2$ and $\log(p(\lambda)/p_\lambda) \leq \theta\eta/4$ on I . Define

$$x_\lambda = X(\lambda)e^{-\theta}, \quad y_\lambda = Y(\lambda)e^{-\theta}, \quad \mu_\lambda = M(\lambda) \quad (\lambda \in I),$$

so $x_\lambda, y_\lambda \in [\eta/2, 1 - \eta]$. By (A1),

$$-\log x_\lambda - t \log y_\lambda \geq \tilde{G}(t) + \theta(1 + t) \quad (t > 0), \quad (16)$$

and by (15), since $1/(1 - \mu_\lambda) \leq 1/\eta$,

$$x_\lambda \leq (1 - \mu_\lambda)p_\lambda^{1/(1-\mu_\lambda)} \exp\left(\frac{\log(p(\lambda)/p_\lambda)}{1 - \mu_\lambda} - \theta\right) \leq (1 - \mu_\lambda)p_\lambda^{1/(1-\mu_\lambda)} e^{-3\theta/4}.$$

By Lemma 3.3 (with $\eta/2$) fix an integer $r \geq 2$ with $(p_\lambda^{1/r} - \mu_\lambda)^r (1 - \mu_\lambda)^{1-r} \geq (1 - \mu_\lambda)p_\lambda^{1/(1-\mu_\lambda)} e^{-\theta/4}$ for all $\lambda \in I$. Then

$$x_\lambda^{1/r} (1 - \mu_\lambda)^{1-1/r} \leq e^{-\theta/(2r)} (p_\lambda^{1/r} - \mu_\lambda) \quad (\lambda \in I), \quad (17)$$

so in particular $p_\lambda^{1/r} - \mu_\lambda \geq x_\lambda^{1/r}(1 - \mu_\lambda)^{1-1/r} \geq \eta/2$. Now fix $\varepsilon_1 > 0$ with

$$(a) \ 2\varepsilon_1 + \frac{4\varepsilon_1}{\eta} \leq (1 - e^{-\theta/(2r)})\frac{\eta}{2}, \quad (b) \ \varepsilon_1 \leq \frac{\eta^3}{4}, \quad (c) \ \varepsilon_1 \leq \frac{\theta\eta}{8}, \quad (d) \ \varepsilon_1 \leq \frac{\eta}{32}.$$

We check the hypotheses of Lemma 3.4 for $(x_\lambda, y_\lambda, \mu_\lambda, p_\lambda)$ with ε_1 in place of ε and $\eta/4$ in place of η , uniformly in $\lambda \in I$. Clearly $\varepsilon_1 < \eta/16$, $\mu_\lambda \geq \eta/4$, $p_\lambda - \varepsilon_1 \geq \eta/4$, and $x_\lambda + \varepsilon_1, y_\lambda + \varepsilon_1, \mu_\lambda + \varepsilon_1 < 1$. For (4), by (17) and (a),

$$x_\lambda^{1/r}(1 - \mu_\lambda - \varepsilon_1)^{1-1/r} + \mu_\lambda + 2\varepsilon_1 \leq p_\lambda^{1/r} - (1 - e^{-\theta/(2r)})\frac{\eta}{2} + 2\varepsilon_1 \leq p_\lambda^{1/r} - \frac{4\varepsilon_1}{\eta} \leq (p_\lambda - \varepsilon_1)^{1/r},$$

the last step because $\frac{d}{du}u^{1/r} = \frac{1}{r}u^{1/r-1} \leq u^{-1} \leq 4/\eta$ on $[\eta/4, 1]$. For (5): $p < 1$ and $1/(1-M) \geq 1$ give $X \leq (1-M)p$, so $1 - \mu_\lambda - x_\lambda \geq (1-M)(1-p) \geq \eta^2$ and $\varepsilon_1(\mu_\lambda + x_\lambda) \leq 2\varepsilon_1 \leq \eta^3/2 \leq \mu_\lambda(1 - \mu_\lambda - x_\lambda)$ by (b). Let $L_0 = L_0(\varepsilon_1, r, \eta/4)$ be as in Lemma 3.4.

The Ramsey bounds consumed by (H1). Let N_G be such that (3) holds for G with $\theta/2$ in place of ε whenever $k + \ell \geq N_G$, and let $L_1 = \max\{L_0, N_G\}$. Let $\ell \geq L_1$, $\lambda \in I$ and $k' \geq 1$ be arbitrary, and $t = \ell/k'$. By (c), $\log(x_\lambda + \varepsilon_1) \leq \log x_\lambda + \varepsilon_1/x_\lambda \leq \log x_\lambda + \theta/4$, and likewise for y_λ ; with (16),

$$-k' \log(x_\lambda + \varepsilon_1) - \ell \log(y_\lambda + \varepsilon_1) \geq k'(\tilde{G}(t) + \frac{3}{4}\theta(1+t)) \geq \tilde{G}(t)k' + \frac{\theta}{2} \max(k', \ell) \geq \log R(k', \ell),$$

where the last inequality is (3) ($k' + \ell \geq N_G$). Thus (H1) holds for every k .

The induction. Let $K \geq \max\{L_1/\varepsilon_0, 2/\varepsilon_0\}$ be so large that for all $k \geq K$: (i) $|F'(s) - F'(s')| \leq \theta'/2$ whenever $s, s' \in [\varepsilon_0/2, 1]$, $|s - s'| \leq 1/k$; (ii) $e^{-\varepsilon k} \leq 1 - e^{-\theta'/2}$ (recall $F \geq 0$); (iii) $2\varepsilon k \geq r \log(2k/\varepsilon_1) + \log 16$; (iv) $e^{(F(\varepsilon_0) + \varepsilon)k} \geq 4$; (v) $e^{\varepsilon k} \geq 2$. Fix $k \geq K$. We prove by induction on $\ell \in \{1, \dots, k\}$ that every coloring of K_N with $N := \lceil e^{(F(\ell/k) + \varepsilon)k} \rceil$ contains a red K_k or a blue K_ℓ . For $\ell \leq \varepsilon_0 k$ this was shown above. Let $\ell > \varepsilon_0 k$, $\lambda = \ell/k \in I$, and consider a coloring of K_N .

Low red density. If the red density is $< p_\lambda$, some vertex has blue degree at least $(1 - p_\lambda)(N - 1) = e^{-F'(\lambda) + \theta'}(N - 1) \geq e^{(F(\lambda) + \varepsilon)k - F'(\lambda) + \theta'/2}$ by (ii). By the mean value theorem, $F(\frac{\ell-1}{k})k = F(\lambda)k - F'(\xi)$ with $\xi \in [\lambda - \frac{1}{k}, \lambda] \subseteq [\varepsilon_0/2, 1]$, and $F'(\xi) \geq F'(\lambda) - \theta'/2$ by (i). Hence the blue neighborhood has at least $\lceil e^{(F((\ell-1)/k) + \varepsilon)k} \rceil$ vertices and, by the induction hypothesis (or the small-ratio case), contains a red K_k or a blue $K_{\ell-1}$, which with the vertex gives a blue K_ℓ .

High red density. If the red density is $\geq p_\lambda$, choose disjoint X, Y with $|X| = |Y| = \lfloor N/2 \rfloor$ maximising $e_R(X, Y)$. For a uniformly random such pair, each edge lies between X and Y with probability $|X||Y|/\binom{N}{2}$, so $\mathbb{E} d(X, Y)$ equals the red density and $d(X, Y) \geq p_\lambda$. Apply Lemma 3.4 with parameters $(x_\lambda, y_\lambda, \mu_\lambda, p_\lambda, \varepsilon_1, r)$, this $\ell \geq \varepsilon_0 k \geq L_1$, and $t = \ell$. (H1) was verified. For (H2): $\alpha_0 = d(X, Y) + \varepsilon_1/(k + \ell) - p_\lambda \geq \varepsilon_1/(2k) > 0$, and by the definitions of x_λ, y_λ , the definition of ψ , and $\theta(1 + \lambda) \leq 2\theta \leq \psi_{\min} \leq 2\psi(\lambda)$,

$$x_\lambda^{-k} y_\lambda^{-\ell} \mu_\lambda^{-\ell} = \exp(k[-\log X - \lambda \log Y - \lambda \log M + \theta(1 + \lambda)]) = \exp(k[2F(\lambda) - 2\psi(\lambda) + \theta(1 + \lambda)]) \leq e^{2F(\lambda)k},$$

while $\alpha_0^r |X||Y| \geq (\varepsilon_1/2k)^r (N/2 - 1)^2 \geq (\varepsilon_1/2k)^r e^{2(F(\lambda) + \varepsilon)k}/16 \geq e^{2F(\lambda)k}$ by (iv) and (iii). Lemma 3.4 shows that (X, Y) is (k, ℓ, ℓ) -good, i.e. the coloring contains a red K_k or a blue K_ℓ .

Consequently $R(k, \ell) \leq e^{(F(\ell/k) + \varepsilon)k} + 1 \leq e^{(F(\ell/k) + 2\varepsilon)k}$ for all $k \geq K$, $\ell \leq k$, by (v). $\square$

4.2 Self-consistency

Theorem 4.2 (self-consistent bootstrap). *Let $F: (0, 1] \rightarrow [0, \infty)$ be C^1 with $F' > 0$ and $C_F < \infty$, and let $M, X, Y: (0, 1] \rightarrow (0, 1)$ be continuous with $X(\lambda) = (1 - M(\lambda))p(\lambda)^{1/(1-M(\lambda))}$, $p = 1 - e^{-F'}$, such that for all $\lambda \in (0, 1]$:*

(B1) $(X(\lambda), Y(\lambda))$ is F -admissible: $-\log X(\lambda) - t \log Y(\lambda) \geq \tilde{F}(t)$ for all $t > 0$;

(B2) $\psi(\lambda) = F(\lambda) + \frac{1}{2}(\log X(\lambda) + \lambda \log M(\lambda) + \lambda \log Y(\lambda)) > 0$.

Then F is valid.

Proof. For $\sigma \geq 0$ put $F_\sigma(\lambda) = F(\lambda) + \sigma(1 + \lambda)$, $X_\sigma = Xe^{-\sigma}$, $Y_\sigma = Ye^{-\sigma}$. We check the hypotheses of Theorem 4.1 for the function $F_{\sigma/2}$, the valid input $G = F_\sigma$ and the triple (M, X_σ, Y_σ) .

- $F_{\sigma/2}$ is C^1 , ≥ 0 , $C_{F_{\sigma/2}} \leq C_F$, and $F'_{\sigma/2} = F' + \sigma/2 > 0$, so $p_{\sigma/2} := 1 - e^{-F' - \sigma/2} \geq p$ and $X_\sigma \leq X = (1 - M)p^{1/(1-M)} \leq (1 - M)p_{\sigma/2}^{1/(1-M)}$, which is (15).
- The symmetric extension of F_σ is $\tilde{F}_\sigma(t) = \tilde{F}(t) + \sigma(1 + t)$ for all $t > 0$ (for $t > 1$: $t F_\sigma(1/t) = tF(1/t) + \sigma(t + 1)$). Hence, by (B1), $-\log X_\sigma - t \log Y_\sigma = -\log X - t \log Y + \sigma(1 + t) \geq \tilde{F}_\sigma(t)$: the pair (X_σ, Y_σ) is F_σ -admissible, which is (A1).
- (A2) for $F_{\sigma/2}$ with (X_σ, Y_σ) reads $F + \frac{\sigma}{2}(1 + \lambda) + \frac{1}{2}(\log X - \sigma + \lambda \log M + \lambda \log Y - \lambda \sigma) = \psi(\lambda) > 0$.

Thus, by Theorem 4.1, if F_σ is valid then $F_{\sigma/2}$ is valid. Since $F \geq 0$, $F_{\log 4} \geq \log 4$, and the constant $\log 4$ is valid ($R(k, \ell) \leq \binom{k+\ell-2}{k-1} \leq 4^k$); so $F_{\log 4}$ is valid, and by induction so is F_{σ_j} with $\sigma_j = 2^{-j} \log 4$ for every j . Given $\varepsilon > 0$ choose j with $2\sigma_j \leq \varepsilon/2$; as $F_{\sigma_j} \leq F + \varepsilon/2$ on $(0, 1]$, validity of F_{σ_j} with $\varepsilon/2$ gives validity of F with ε . $\square$

Remark 4.3. Theorem 4.2 will be applied, in Section 5, to data that are only piecewise continuous. Its proof (through Theorem 4.1) uses the continuity of M, X, Y and of ψ only to produce, on each compact $I = [\varepsilon_0, 1]$, uniform bounds $X, Y, M, p \in [\eta, 1 - \eta]$ and $\inf_I \psi > 0$, and it uses the identity for X only through the inequality (15). The theorem therefore holds verbatim with “continuous” replaced by these two conclusions—in particular for the piecewise-quadratic C^1 spline F of Section 5, whose X and Y are step functions with finitely many values and whose per-cell slack bounds are part of the certificate. The formalized ladder theorem (Remark 4.4) is stated in exactly this weakened form and is applied unchanged. Self-consistency is also what legitimises the tail of the certificate: below λ_0 the input bound against which admissibility is tested is F itself, not the Erdős–Szekeres bound, so F may simply be continued by a straight line (Section 5.2).

Remark 4.4. Theorem 4.2 has been formalized in Lean 4 on top of the RamseyLean development accompanying [4] (which formalizes Lemma 3.4 and Theorem 4.1 as `Candidate.isGood_of_density_card_1` and `uniformRamseyExpBound_of_descent`): file `SelfConsistent.lean`, theorem `uniformRamseyExpBound_self` depending only on the axioms `propext`, `Classical.choice`, `Quot.sound`. The only change to the existing development is the relaxation of the identity for X in their Theorem 14 to the inequality (15), whose proof already used only the inequality. The proof above is the formalized one; the shift $\sigma(1 + \lambda)$ (rather than a constant) is what makes every rung need only the pointwise slack $\psi > 0$.

Since the first version of this paper, the *entire* proof of Theorem 1.1 has been formalized; for the present revision the development is the directory `RamseyLean/Bootstrap2` of the artifact. The rate function is *defined from the certificate data*: D and M are the piecewise-linear interpolants of the nodes (clamped outside $[\lambda_0, 1]$) and $F := F_0 + \int_{\lambda_0}^\cdot D$, so that `HasDerivAt F (D r)` r is the fundamental theorem of calculus, and the node values $F(\lambda_j) = F_j$ are exact by the integer chain identity (19). A Bool-valued cell checker (integer interval arithmetic at scale 10^{12} , `exp/log` enclosed by Taylor series with directed rounding), mirrored bit-exactly by the Python verifier of Section 5.3, is proved sound against the real-valued definitions; the validity of the tangent witnesses (each is literally a node of the certificate) is established by a monotone two-pointer merge pass; and the ladder theorem of the preceding paragraph is applied unchanged (Remark 4.3). Every cell is discharged by the Lean *kernel* (`decide +kernel`; `native_decide` is not used): 35 770 cells at 0.3–0.6 s per cell (Apple M-series / AMD Genoa), about six core-hours over 559 chunk files of at most 64 cells each; a checking process needs about 1.5 GB, mostly the import closure (against 5.3 GB per 48-cell chunk for the ansatz certificate of the previous revision). The final theorems, each depending only on the axioms `propext`, `Classical.choice`, `Quot.sound`, are `main_bootstrap2` (validity of the spline rate function), `bootstrap2_diagonal` ($R(k, k) \leq e^{(F_N + \varepsilon)k}$ eventually, for every $\varepsilon > 0$, with the explicit rational F_N of Theorem 1.1), and `bootstrap2_diagonal_num`: $R(k, k) \leq (37690/10000)^k$ for all large k —a fully formal decimal bound, where the previous revision stated the decimal form only in prose; the sharp enclosure $e^{F_N} \leq 37689719/10^7$ is likewise a kernel theorem (`exp_FN_le_sharp`).

Remark 4.5. The ladder is what makes it legitimate to feed the bound F into its own admissible region. A direct induction on $k + \ell$ seems to fail: the region hypothesis is needed for pairs (k', ℓ) with $k' < k$ and the same ℓ , whose sums may fall below any fixed starting point of the induction, and the trivial bound $R \leq 4^{k+\ell}$ at the base is exponentially too weak to be absorbed.

5 The maximal admissible Y and the spline certificate

5.1 A Legendre description of the admissible boundary

For fixed x , the largest y with (x, y) F -admissible is $y = e^{-S}$ with $S = \sup_{t>0} (\tilde{F}(t) + \log x)/t$. When $\tilde{F}$ is concave the supremum is located by a tangent line; this is the content of [4, Lemma 15], in the following form.

Lemma 5.1. *Let $\varphi: (0, \infty) \rightarrow \mathbb{R}$ be concave, $u < 0$, and $S(u) = \sup_{t>0} (\varphi(t) + u)/t$. If $t_0 > 0$ and σ is a supergradient of φ at t_0 (i.e. $\varphi(t) \leq \varphi(t_0) + \sigma(t - t_0)$ for all t) with $\varphi(t_0) - \sigma t_0 + u \leq 0$, then $S(u) \leq \sigma$.*

Proof. $(\varphi(t) + u)/t \leq (\varphi(t_0) - \sigma t_0 + u)/t + \sigma \leq \sigma$ for every $t > 0$. □

Suppose $F: (0, 1] \rightarrow (0, \infty)$ is concave and C^1 with $F' > 0$, and let $M, X: (0, 1] \rightarrow (0, 1)$ be given. Put

$$S(\lambda) = \sup_{t>0} \frac{\tilde{F}(t) + \log X(\lambda)}{t}, \quad Y(\lambda) = e^{-S(\lambda)}, \quad \psi(\lambda) = F(\lambda) + \frac{1}{2}(\log X + \lambda \log M - \lambda S)(\lambda), \quad (18)$$

so that $Y(\lambda)$ is the largest y for which $(X(\lambda), y)$ is F -admissible, and ψ is the slack in (B2) for this maximal choice; when X is the extremal value $(1 - M)p^{1/(1-M)}$, $p = 1 - e^{-F'}$, of Theorem 4.2,

(18) is the slack of the self-consistent system for given F and M . Admissibility of a pair (x, y) splits into the two orientations

$$(a) \quad -\log x - s \log y \geq F(s) \quad (0 < s \leq 1), \quad (b) \quad -\log y - s \log x \geq F(s) \quad (0 < s \leq 1),$$

(b) being the condition at ratios $t = 1/s \geq 1$, where $\tilde{F}(t) = tF(1/t)$. Each orientation is certified by one tangent line: if $u \in (0, 1]$ then, by concavity, $F(s) \leq F(u) + (s - u)F'(u)$ for all $s \in (0, 1]$, and the inequality $F(u) + (s - u)F'(u) \leq -\log x - s \log y$ is *linear* in s , so its validity at $s = 0$ and $s = 1$ implies (a); this is Lemma 5.1 with $\varphi = F$, witness $t_0 = u$ and $\sigma = F'(u)$. Likewise for (b), with x and y interchanged. In contrast with the closed-form ansatz of the previous revision of this paper, no concavity of $\tilde{F}$ across the kink at $t = 1$ is required: the two orientations are certified separately, each inside $(0, 1]$.

5.2 The spline certificate

The proof object replaces the explicit functions of the previous revision by *free data*: a list of $N = 35\,770$ cells covering $[\lambda_0, 1]$, $\lambda_0 = 10^{-4}$ (a subdivision of the 10 560-interval partition on which the fixed point was computed; Section 5.3). Cell j ($0 \leq j < N$) carries two *nodes* $(\lambda_j, F_j, d_j, M_j)$ and $(\lambda_{j+1}, F_{j+1}, d_{j+1}, M_{j+1})$ —consecutive cells share a node—two constants X_j, Y_j , and two tangent *witnesses* w_j^a, w_j^b , each of which is literally one of the $N + 1$ nodes of the certificate. All entries are exact scaled integers: λ, d, M, X, Y are integer multiples of 10^{-12} and F of $\frac{1}{2} \cdot 10^{-24}$, so that the trapezoid identity

$$F_{j+1} = F_j + \frac{1}{2} (d_j + d_{j+1})(\lambda_{j+1} - \lambda_j) \quad (19)$$

is *exact* integer arithmetic (the factor $\frac{1}{2}$ is absorbed by the scale of F).

From the data define D and M as the piecewise-linear interpolants of (λ_j, d_j) and (λ_j, M_j) , clamped to the constants d_0, M_0 on $(0, \lambda_0]$; let

$$F(r) = F_0 + \int_{\lambda_0}^r D(t) dt,$$

and let X, Y be the step functions with values X_j, Y_j on cell j , frozen at X_0, Y_0 on $(0, \lambda_0]$. Then F is C^1 and piecewise quadratic with $F' = D > 0$, concave because D is non-increasing, linear on the tail $(0, \lambda_0]$, and $F(\lambda_j) = F_j$ *exactly*, by (19).

Proposition 5.2 (the certificate). *The file `pq_cert_final.json` identified in Section 6 consists of 35 770 cells as above, with $\lambda_0 = 10^{-4}$ and $\lambda_N = 1$. For every cell j , with $\mathcal{D}_j = [d_{j+1}, d_j]$ and $\mathcal{M}_j = [M_j, M_{j+1}]$:*

(C0) *structure*: $0 < d_{j+1} \leq d_j$; $0 < M_j \leq M_{j+1} < 1$; $0 < X_j, Y_j < 1$; and the chain identity (19);

(C1) *the X-inequality*: $X_j \leq (1 - e^{-D})^{1/(1-M)}(1 - M)$ for all $(D, M) \in \mathcal{D}_j \times \mathcal{M}_j$;

(C2) *admissibility*: the witness w_j^a is a node u with (its node values $F(u), D(u)$ being exact data)

$$F(u) - u D(u) \leq -\log X_j \quad \text{and} \quad F(u) + (1 - u) D(u) \leq -\log X_j - \log Y_j,$$

and the witness w_j^b satisfies the same two inequalities with X_j and Y_j interchanged;

(C3) *slack*: with $\hat{\psi}_i = F_i + \frac{1}{2}(\log X_j + \lambda_i \log M_i + \lambda_i \log Y_j)$ for $i \in \{j, j+1\}$ and the dip bound $\delta_j = \frac{1}{8}(\lambda_{j+1} - \lambda_j)(\log M_{j+1} - \log M_j) \geq 0$: $\hat{\psi}_j > \delta_j$ and $\hat{\psi}_{j+1} > \delta_j$.

Moreover the witnesses w_j^a are non-decreasing and the w_j^b non-increasing in j , each one a node of the certificate; and the single tail inequality

$$(T) \quad F_0 - d_0 \lambda_0 + \frac{1}{2} \log X_0 > 0$$

holds. Finally $F_N = 2653604512524788171208898/(2 \cdot 10^{24})$ and $e^{F_N} \leq 3.7689719$.

Proof. By computation, in interval arithmetic at scale 10^{-12} with exp and log enclosed by Taylor series with directed rounding, every comparison being between integers: once by the Lean kernel and once by a bit-exact Python mirror of the Lean checker (Section 5.3). $\square$

Proposition 5.2 implies Theorem 1.1, through Theorem 4.2 in the form of Remark 4.3. First the global hypotheses: $F' = D > 0$; $F(0^+) = F_0 - d_0 \lambda_0 > 0$ by (T), as $\log X_0 < 0$, so $F > 0$; and $C_F < \infty$ because $(H(\lambda) - F(\lambda))/\lambda \leq \log(1/\lambda) + 1 - F(0^+)/\lambda \rightarrow -\infty$ as $\lambda \downarrow 0$. Now fix r in cell j . The interpolants satisfy $D(r) \in \mathcal{D}_j$ and $M(r) \in \mathcal{M}_j$, so (C1) gives (15) for the step value X_j . For (B1): by concavity, $F(s) \leq F(u) + (s - u)D(u)$ for every node u and all $s \in (0, 1]$, so the two endpoint checks (C2), linear interpolation in s , and the tangent bound give, for all $s \in (0, 1]$,

$$F(s) \leq -\log X_j - s \log Y_j \quad \text{and} \quad F(s) \leq -\log Y_j - s \log X_j,$$

i.e. $-\log X_j - t \log Y_j \geq \tilde{F}(t)$ for $t \leq 1$ (first inequality at $s = t$) and for $t \geq 1$ (second inequality at $s = 1/t$, multiplied by t): the pair (X_j, Y_j) is F -admissible. For (B2), write $r = (1 - \theta)\lambda_j + \theta\lambda_{j+1}$. Concavity places F above its chord, $F(r) \geq (1 - \theta)F_j + \theta F_{j+1}$; concavity of log places $\log M(r)$ —the argument being the *linear* interpolant of the M_i —above the linear interpolant of the $\log M_i$; and the convex quadratic $r \cdot$ (linear interpolant of the $\log M_i$) lies below the linear interpolant of the $\lambda_i \log M_i$ by exactly $\theta(1 - \theta)(\lambda_{j+1} - \lambda_j)(\log M_{j+1} - \log M_j)$. Hence

$$\psi(r) \geq (1 - \theta)\hat{\psi}_j + \theta\hat{\psi}_{j+1} - \frac{1}{2}\theta(1 - \theta)(\lambda_{j+1} - \lambda_j)(\log M_{j+1} - \log M_j) \geq \min(\hat{\psi}_j, \hat{\psi}_{j+1}) - \delta_j > 0$$

by (C3), since $\theta(1 - \theta) \leq \frac{1}{4}$. This gives (B2) on $[\lambda_0, 1]$. Finally the tail $r \in (0, \lambda_0]$: there D, M, X, Y are frozen at the values d_0, M_0, X_0, Y_0 , which lie in cell 0's boxes, so (15) holds by cell 0's check (C1), and admissibility of (X_0, Y_0) is a global statement already certified by cell 0's (C2); moreover ψ is *linear* on the tail, equal to $\hat{\psi}_0 > 0$ at λ_0 and to $F(0^+) + \frac{1}{2} \log X_0 = F_0 - d_0 \lambda_0 + \frac{1}{2} \log X_0 > 0$ in the limit $\lambda \downarrow 0$, by (T). Theorem 4.2 now gives validity of F , and $F(1) = F_N$ with $e^{F_N} \leq 3.7689719 < 3.769$ is Theorem 1.1.

This is where self-consistency pays a second dividend. The previous revision needed a hand analysis of small λ (its Lemma 5.3) because its F was pinned to a closed form whose behavior as $\lambda \downarrow 0$ had to be estimated; the certificate's F is free data and may simply be *continued by a straight line* below λ_0 , self-consistency making the frozen pair (X_0, Y_0) admissible against the extended F itself, with no appeal to the Erdős–Szekeres bound below λ_0 . The one-line inequality (T) is all that remains of the small- λ regime.

5.3 The computer-assisted part

Three computations occur in this paper, with different status.

(a) A *heuristic* floating-point computation (`opt.py`, `opt2.py`) discovers the data; it plays no role in the proof. It computes the fixed point of the bootstrap in the Theorem-13 discretization of [4]: on each interval of a partition of $[\lambda_0, 1]$, greedily from the left, the minimal slope d admissible for the current F —tested in *both* orientations, self-consistently, against $U = F + \eta$ —together with a per-interval book parameter M and pair (X, Y) ; the sweep is iterated to convergence. The mesh sequence 600, 1200, 2400, 4800, 9600 gives $e^{F(1)} = 3.77003, 3.76943, 3.76912, 3.76897, 3.768894$: clean first-order convergence, with Richardson limit ≈ 3.76881 , so little is left on the table within this framework.

(b) The *proof object* is the exact scaled-integer certificate of Proposition 5.2, exported by `pq_export.py` from the converged fixed point on the 10 560-interval partition: nodal slopes and M are rounded to the integer scales; the F_j are recomputed by the exact chain (19); the X_j, Y_j are rounded down; cells are subdivided (to the final 35 770) until every interval-arithmetic check passes with margin (admissibility margin $6 \cdot 10^{-7}$ per cell, slack margin 10^{-6}); and the base value F_0 is raised by $2.08 \cdot 10^{-5}$ so that the tail inequality (T) holds. The certified constant exceeds the raw fixed point by $7.8 \cdot 10^{-5}$ in total—the price of rounding, of freezing (X, Y) per cell, and of the margins.

(c) An *independent check*: `verify_pq.py`, a bit-exact Python mirror of the Lean checker (on top of `fpi.py`, an op-for-op port of the Lean fixed-point interval arithmetic), re-verifies the chain, every cell, the witness merge and the tail in about two minutes. The strongest check, however, is the Lean kernel itself (Remark 4.4): the formal development is self-contained and trusts none of the Python computations.

The certified terminal value is the exact rational $F_N = 2653604512524788171208898 / (2 \cdot 10^{24}) = 1.32680225\dots$, and the kernel-checked enclosure (exp of the halved argument by a degree-16 Taylor series with directed rounding, then squared) gives $e^{F_N} \leq 3.7689719 < 3.769$.

6 Both orientations, and two remarks

Both orientations

Admissibility is a requirement at *every* ratio $t \in (0, \infty)$: the pair (x, y) must dominate the Ramsey bound both for $\ell \leq k$ and for $\ell > k$. Lemma 15 of [4], which coincides with Lemma 5.1 above, handles both orientations, and in the certificate of Section 5 the point is structural: every cell carries one tangent witness *per orientation*, so both orientations are tested for every cell. As a consistency check, the self-consistent framework recovers the constant of [4] directly: with $F = H + G$ the function of (1) and $M = \lambda e^{-\lambda}$, the self-consistent slack (18) satisfies $\psi(1) = 6.1 \cdot 10^{-4} > 0$ and, numerically, $\psi > 0$ on $(0, 1]$, so Theorem 4.2 re-proves (1) without any chain of iterations.

A proposed quintic correction

Appendix A.2 of [5] reports the function $F = H + e^{-\lambda}(-0.25\lambda + 0.033\lambda^2 + 0.08\lambda^3 - 0.0778\lambda^5)$, which would give $e^{F(1)} \approx 3.6961$, obtained by an automated search within the framework of [4]. For this F , with Y the maximal F -admissible choice (18) and M optimized pointwise, one finds $\psi(1) = -0.094$ and $\psi < 0$ on $[0.6, 1]$: the function is not self-consistent, and no round of the bootstrap whose input bound dominates F pointwise (in particular none starting from (1)) can produce it. This is consistent with the expectation in [4, Remark 17] that constants below 3.75 would require new ideas.

Possible improvements

Within the present framework the remaining gap is pure discretization: the mesh sequence of Section 5.3 converges at first order to ≈ 3.76881 , about 10^{-4} below the certified constant, so little is left. It is worth recording that the two techniques of this paper subsume each other’s gains. Imposing the self-consistent linear tail on the discretized fixed point changes its value by less than 10^{-8} —the greedy slope recursion is strongly attracting in the base value. And restricting the admissibility requirement to the ratios $t \geq \lambda$ actually consumed by (H1) (the pairs (k', ℓ) needed there have $k' \leq k$) changes nothing: the binding ratio is $\approx 3.8\lambda$ throughout. Genuinely better constants would have to come from the combinatorial part—e.g. a sharper key lemma.

Code and data

All code is at github.com/wamlat/RamseyLean-bootstrap and archived on Zenodo under the concept DOI 10.5281/zenodo.22263823, which resolves to the latest archived version (this revision: tag v2.0.2, commit of record 23572c0 for the three final theorems; the previous 3.772 revision is version DOI 10.5281/zenodo.22263824); the artifact is identified by the SHA-256 hashes below. The Python verifier uses only the standard library. Reproduction:

```
python3 verify_pq.py pq_cert_final.json    bit-exact mirror of the Lean checker, about 2 min
lake build RamseyLean.Bootstrap2.CertMain  kernel verification, about six core-hours
```

(the latter preceded by `lake exe cache get` and followed by `#print axioms` on the three final theorems of Remark 4.4). The kernel build checks the 559 chunk files of `Bootstrap2/CertData2` (at most 64 cells each; about 1.5 GB per checking process), and the chunk targets parallelise up to the memory budget; the clean-clone verification of record (`verification/cleanverify-x86-64.log`) took 28 minutes on 16 x86-64 cores.

pq_export.py	0822413e2f8d99dd53ab0ac016ccffce46955068f182cd7c598e9c2e6862d309
verify_pq.py	362731555535c8275aaffa9dafd3b08fa6d104c7e39f7ee176c4f6023ca83c22
fpi.py	b01c4d07ac271ea74e104c652c639b04e64a99e4729a5b4bd8b1a4851f839f1f
funcs.py	237344ad8502f404573102446d90694695fd4f0e9eb5fdbfb00095ccdd2cf6ef
pq_cert_final.json	462cc882a82b08711347cb2af526ea54a46290f6088d1bf82cce66159d66ee1d
opt.py	0a91be491c4ac87d988c94f784cf27cc988fa9a8edff3ec858b55017690093ff
opt2.py	168fe19d9eb5c6e478a7d2e1b095c1eeaa991620753aed2ecefbd29e431b5caa

The Lean 4 formalization (Remark 4.4) consists of the shift ladder (`SelfConsistent.lean`, unchanged from the previous revision) and the directory `RamseyLean/Bootstrap2/`: the certificate data types and spline definitions, the analytic layer (F from the data by integration, tangent bound, interpolant boxes), the cell checker with its soundness proof, the coverage, witness-merge and tail passes, the glue, and the 35 770-cell certificate data in 559 chunk files.

SelfConsistent.lean	e4b9e7780fc11ecac9e7f516e6e5f074abd60f884a1434269a1baee08b4a6143
Bootstrap2/Defs.lean	e1999435791780a08e655a0436de2858f329a45d9aef5348d9d2fece0ad6e1cb
Bootstrap2/Spline.lean	95017cf347a0e6a969923343187a3dc2a61c370825846ce2acdd573b0f087132
Bootstrap2/Check.lean	a954f59aa2da3cddb88c19c82435b4f252942ef8fc03f847824fe69ce04d4d43f
Bootstrap2/Cover.lean	28eaadb92c83b973816b1526c90ef6fa5e0477342882dbb021885d85bf0d12fe
Bootstrap2/Glue.lean	440000987f5cb4a934fec48eef146041b56295d31efa024391ee6ff5b9cc93f5
Bootstrap2/Main2.lean	7b709fbca384b94c92b8426909f4b119a31e0a81712ade5ac5e3891f0f21e048
Bootstrap2/CertMain.lean	20fd323b9ccfb7f1108bbd9c5e41c067291b7b700a92edceb697d7b76e5e9c74
Bootstrap2/CertData2 (cat Chunk001..559 CertData2.lean)	2ee95b1bc491019c77ec9a4462a8446e76ae64456769948717f61531406d72fa

The repository also retains, unchanged, the artifacts of the previous revision (the Arb [1] certificate, its two checkers, and the directory `lean/Bootstrap/`), and the scripts reproducing the numbers quoted in Section 6.

References

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- [5] *HorizonMath: Measuring AI progress toward mathematical discovery*, arXiv:2603.15617v1 (2026), Appendix A.2.
- [6] N. Ndiaye, *Guarantee of cliques and uncertainty of trees: Ramsey theory and leaf powers*, PhD thesis, McGill University, 2025.